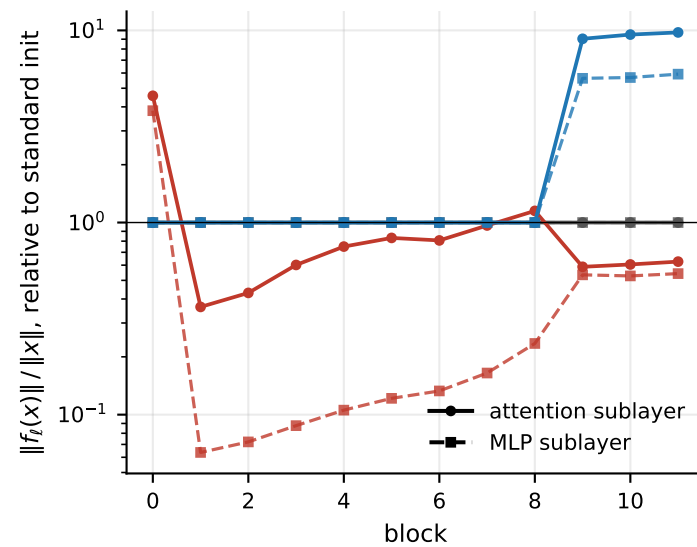
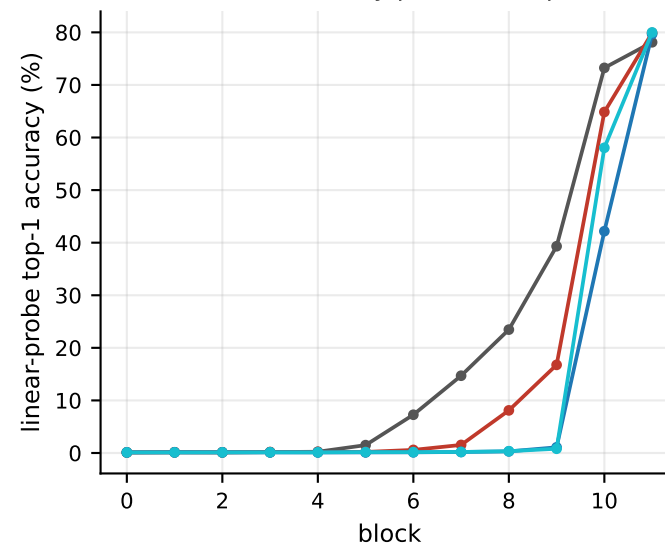


— standard init (timm trunc-normal, std 0.02) — final top-1 78.1%
— early blocks 0-8: per-matrix weight std set to the reference network's (weights zero-mean; q,k $\times 3-4$, v $\times 1.2-2$, MLP $\times 0.7-1.2$ vs standard), LayerNorm gains 0.3-0.5 instead of 1 — final top-1 79.9%
— late blocks 9-11: write matrices (v, proj, fc2) scaled so that $\|f(x)\|/\|x\| = 1.4$ for both sublayers — final top-1 79.7%
— late blocks 9-11: write matrices scaled to the reference network's $\|f(x)\|/\|x\|$ — final top-1 80.0%

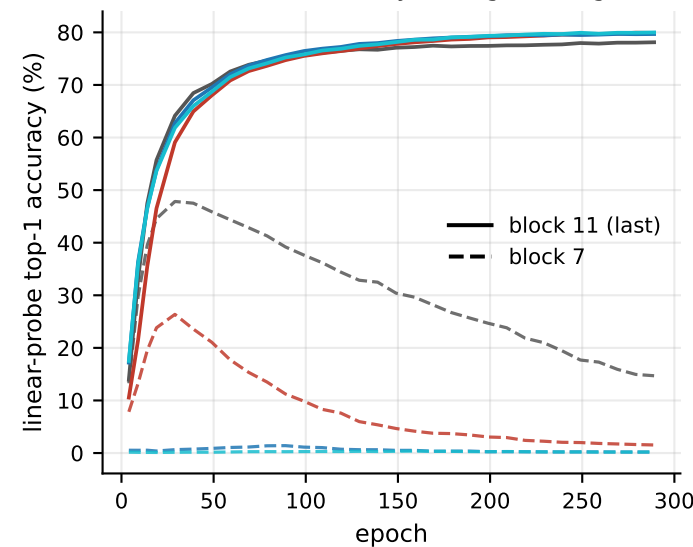
(a) Residual-write ratio at initialisation



(b) Class decodability per block, epoch 289



(c) Probe accuracy during training



(d) Residual-write ratio at epoch 289

